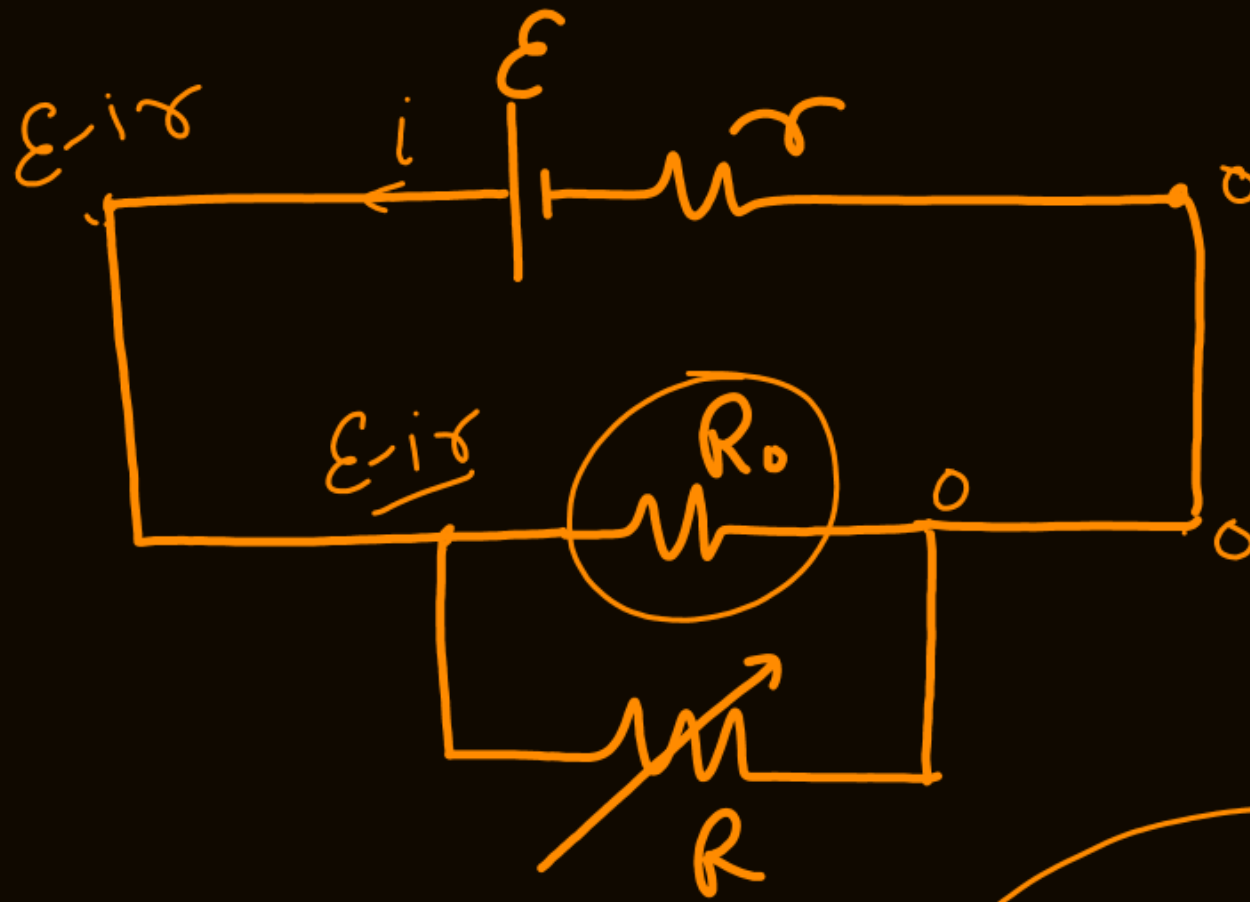
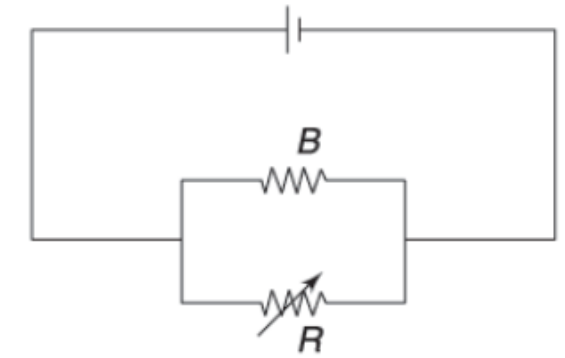


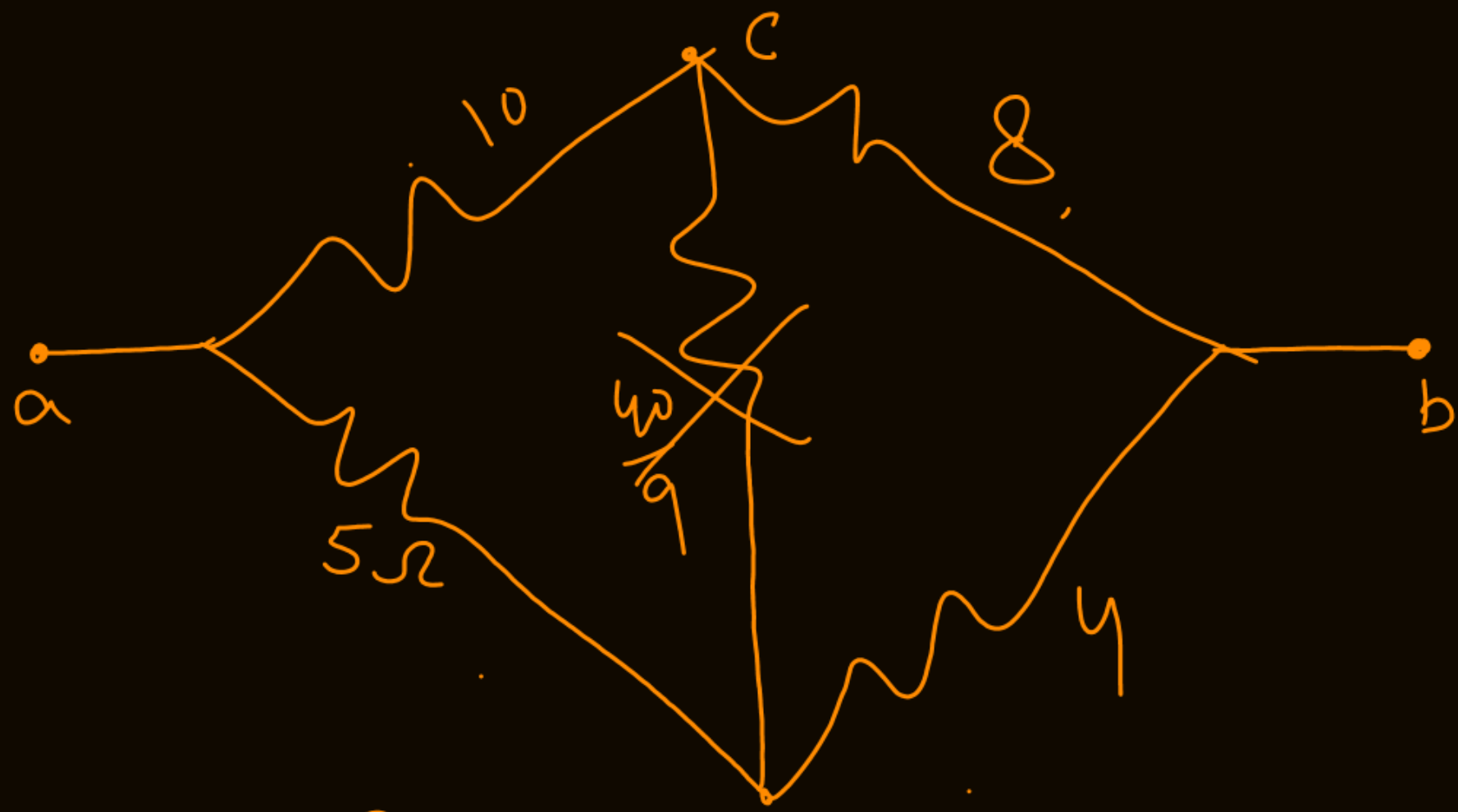


$$\begin{aligned}
 &\downarrow P = \mathcal{E}i \quad i = \frac{\mathcal{E}}{R_{eq}} \quad \downarrow R_{eq} = \frac{R_0 R}{R_0 + R} + r \quad R \uparrow \\
 &P = (\mathcal{E} - ir) R_0
 \end{aligned}$$

Q. 8: A bulb B is connected to a source having constant emf and some internal resistance. A variable resistance R is connected in parallel to the bulb. If the resistance R is increased, how will it affect the—

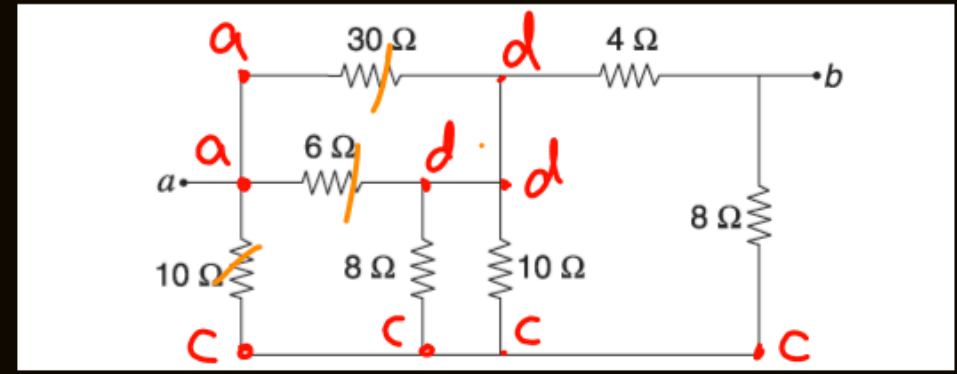
- Brightness of the bulb?
- Power spent by the source?





$$\frac{18 \times 9}{18 + 9} = \frac{18 \times 9}{27} = \underline{\underline{6\Omega}}$$

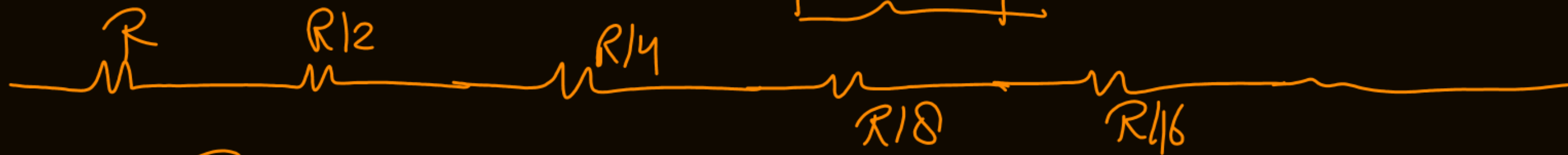
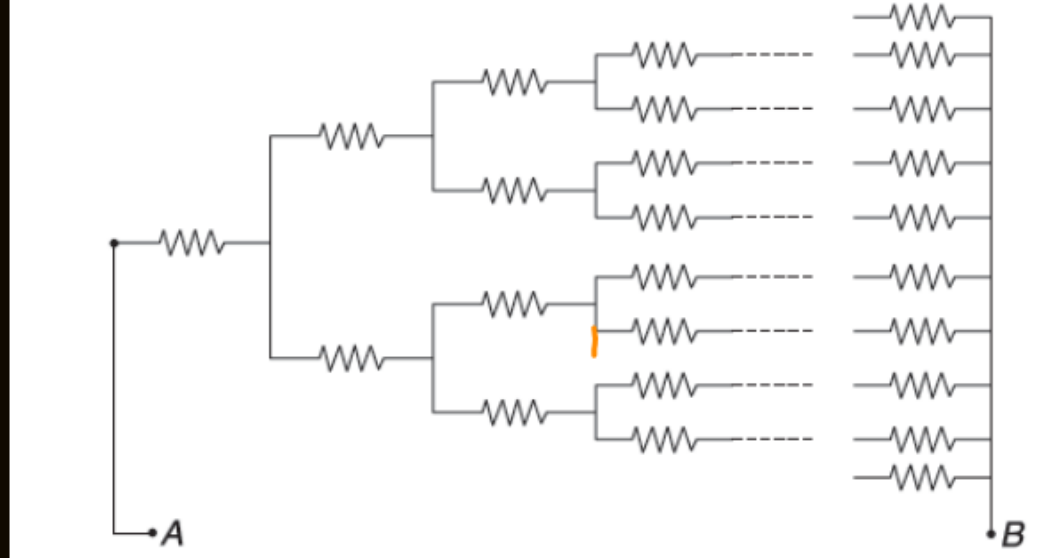
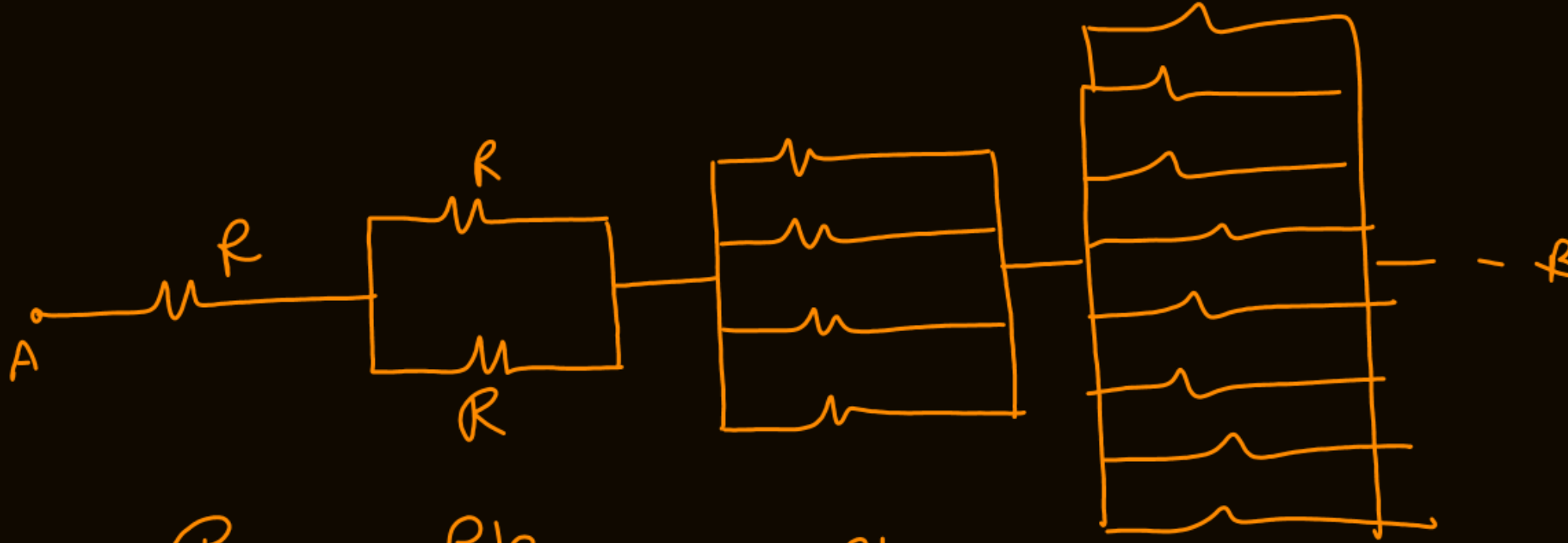
Q. 10: Find the equivalent resistance between point *a* and *b* in the network shown in figure.



$$\frac{30 \times 6}{30 + 6} = 5$$

$$\frac{10 \times 4}{10 + 4} = \frac{40}{14}$$

Q. 12: An infinite network of resistances has been made as shown in the Figure. Each resistance is R . Find the equivalent resistance between A and B .



$$\begin{aligned}
 R_{eq} &= R + R/2 + R/4 + R/8 + R/16 \\
 &= R \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \right) = R \times \frac{1}{1 - 1/2} = 2R
 \end{aligned}$$

$$i_c + i_B = i_e$$

$$i_c + i_B = \frac{i_c}{0.9}$$

$$i_B = i_c \left\{ \frac{10}{9} - 1 \right\}$$

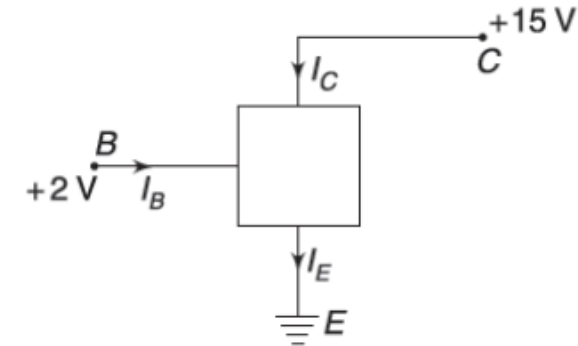
$$= \frac{i_c}{9}$$

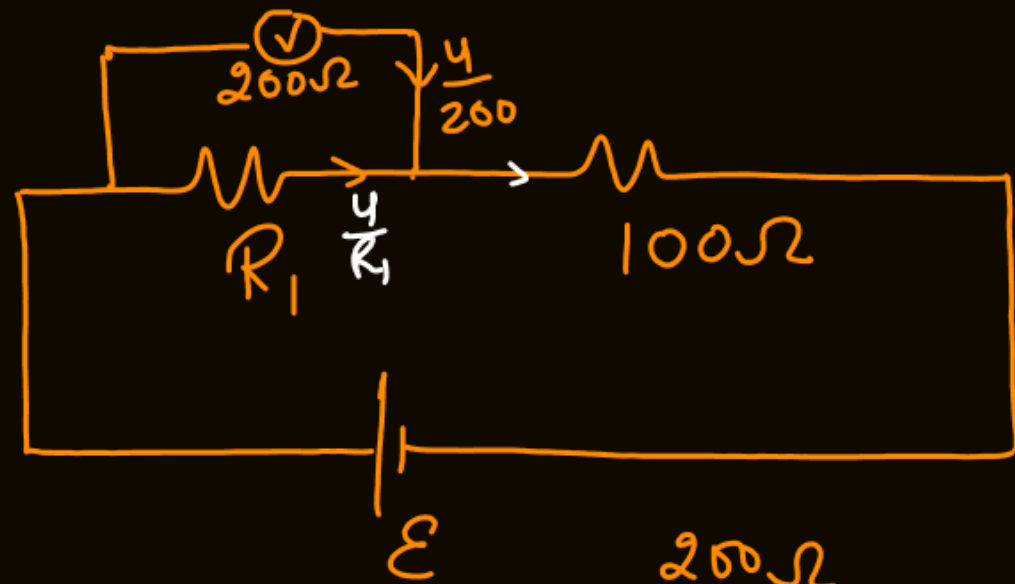
$$i_c = 9 i_B$$

$$\Delta i_c = 9 \Delta i_B$$

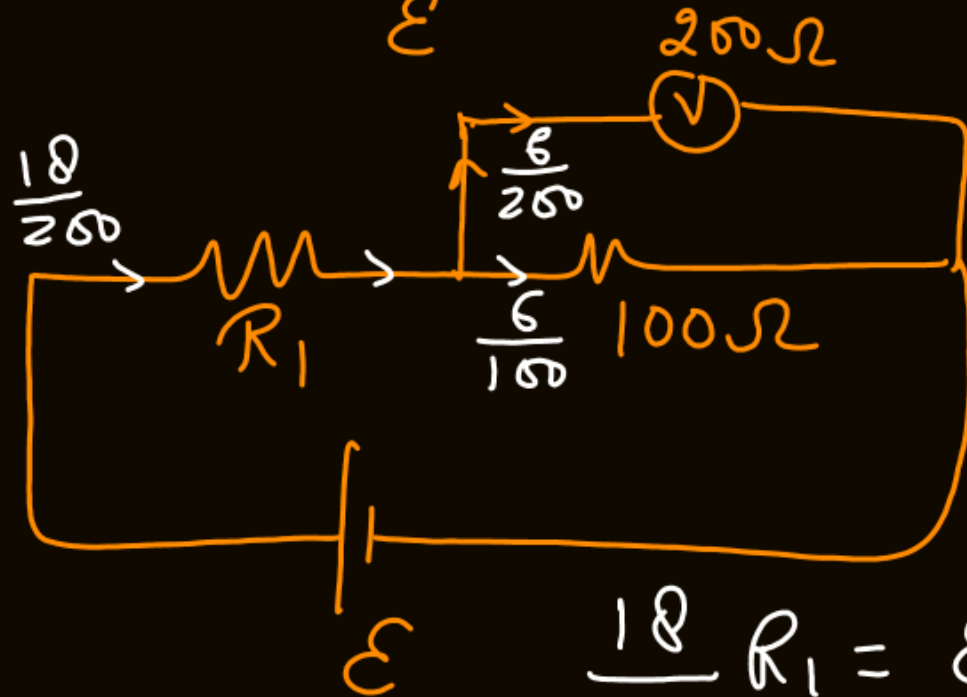
Q. 18: The box shown in the figure has a device which ensures that $I_C = 0.9 I_E$.

If a small change (ΔI_B) is made in I_B , calculate the corresponding change in I_C .





$$\left(\frac{4}{200} + \frac{4}{R_1} \right) \times 100 = \mathcal{E} - 4 \quad \text{--- (1)}$$

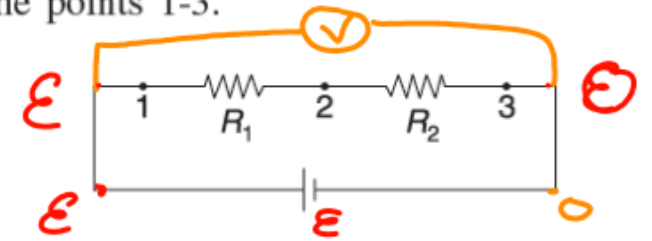


$$\frac{6}{200} + \frac{6}{100} = \frac{18}{200}$$

$$\frac{18}{200} R_1 = \mathcal{E} - 6 \quad \text{--- (2)}$$

$$\mathcal{E} = \frac{18}{200} \times \frac{200}{3} + 6 = 12 = \mathcal{E}$$

Q. 27: In the circuit shown in the Figure, cell is ideal and $R_2 = 100 \, \Omega$. A voltmeter of internal resistance $200 \, \Omega$ reads $V_{12} = 4 \, \text{V}$ and $V_{23} = 6 \, \text{V}$ between the pair of points 1-2 and 2-3 respectively. What will be the reading of the voltmeter between the points 1-3.



$$\Rightarrow \frac{1}{\mathcal{E}}$$

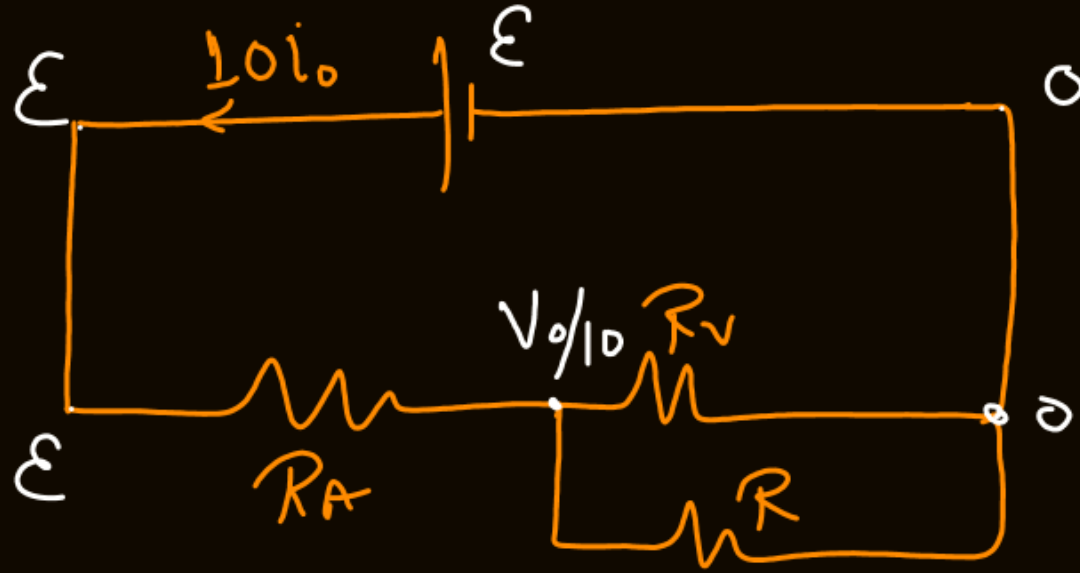
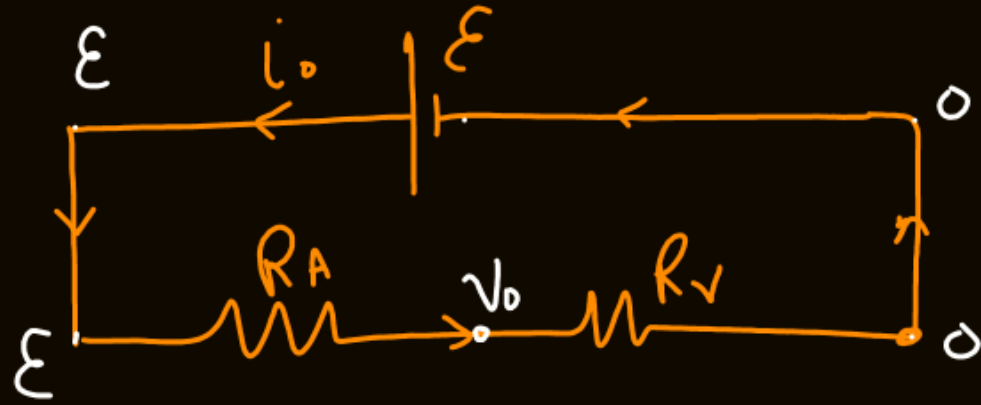
$$2 + \frac{400}{R_1} + 4 = \mathcal{E} = \frac{18R_1}{200} + 6$$

$$18R_1^2 = 200 \times 400$$

$$R_1^2 = \frac{200 \times 200}{9}$$

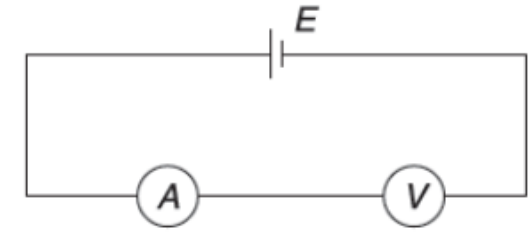
$$R_1 = \frac{200}{3} \, \Omega$$

$$i_0 = \frac{\mathcal{E} - V_0}{R_A}$$



$$i = \frac{\mathcal{E} - \frac{V_0}{10}}{R_A} = 10i_0 = 10 \frac{\mathcal{E} - V_0}{R_A}$$

Q. 28: In the circuit shown, an ideal cell of emf E is connected in series to a non-ideal ammeter and voltmeter. Reading of the voltmeter is V_0 . When a resistance is added in parallel to the voltmeter its reading becomes $\frac{V_0}{10}$ and the reading of the ammeter becomes 10 times the earlier value. Find V_0 in terms of E .



$$\frac{10\mathcal{E} - V_0}{10R_A} = \frac{10\mathcal{E} - V_0}{R_A}$$

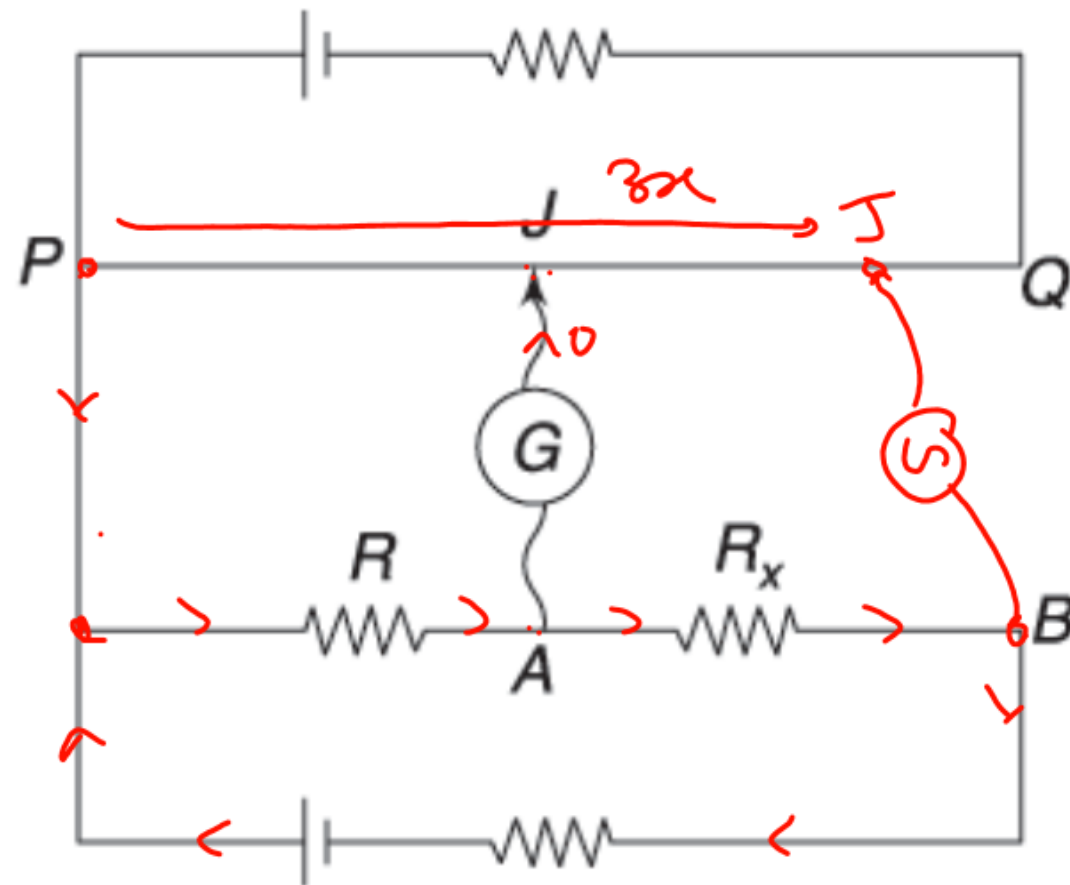
$$10\mathcal{E} - V_0 = 100\mathcal{E} - 100V_0$$

$$90\mathcal{E} = 99V_0$$

$$V_0 = \frac{10\mathcal{E}}{11}$$

Sol:

Q. 32: In the figure shown PQ is a potentiometer wire. When galvanometer is connected at A , it shows zero deflection when $PJ = x$. Now the galvanometer is connected to B and it shows zero deflection when $PJ = 3x$. Find the value of unknown resistance R_x in term of R .



$$iR = kx$$

$$i(R + R_x) = k3x$$

$$R + R_x = 3R$$

$$R_x = 2R$$

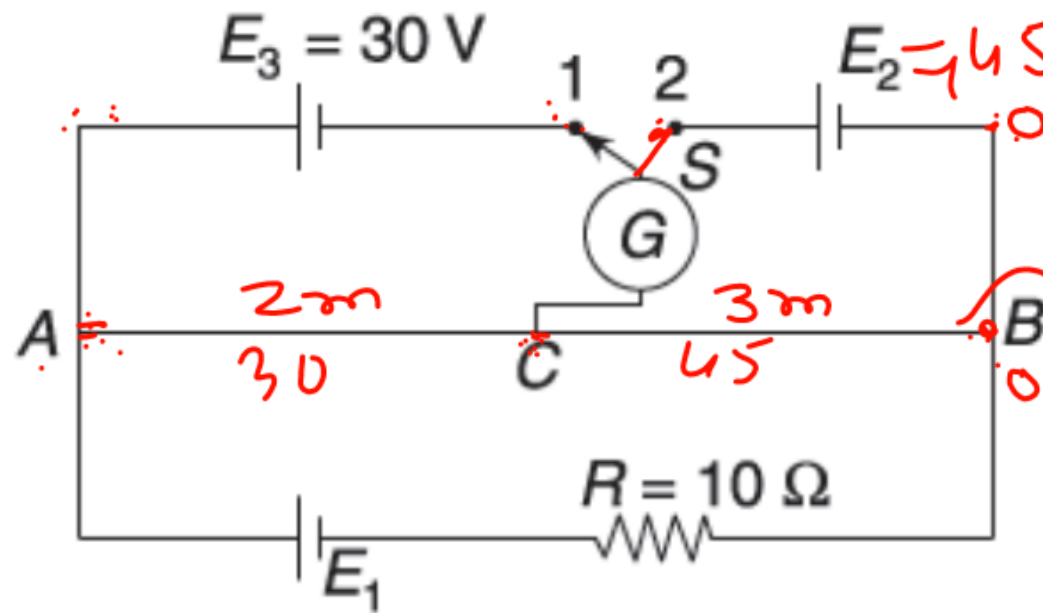
Q. 38: In the circuit shown in Figure AB is a uniform wire of length $L = 5\text{m}$. It has a resistance of $2\ \Omega/\text{m}$. When $AC = 2.0\text{ m}$, it was found that the galvanometer shows zero reading when switch s is placed in either of the two positions 1 or 2. Find the emf E_1 .

$$\mathcal{E}_2 = k \times 3$$

$$30 = k \times 2$$

$$\frac{\mathcal{E}_2}{30} = \frac{3}{2}$$

$$\underline{\underline{\mathcal{E}_2 = 45}}$$



$$\underline{7.5V}$$

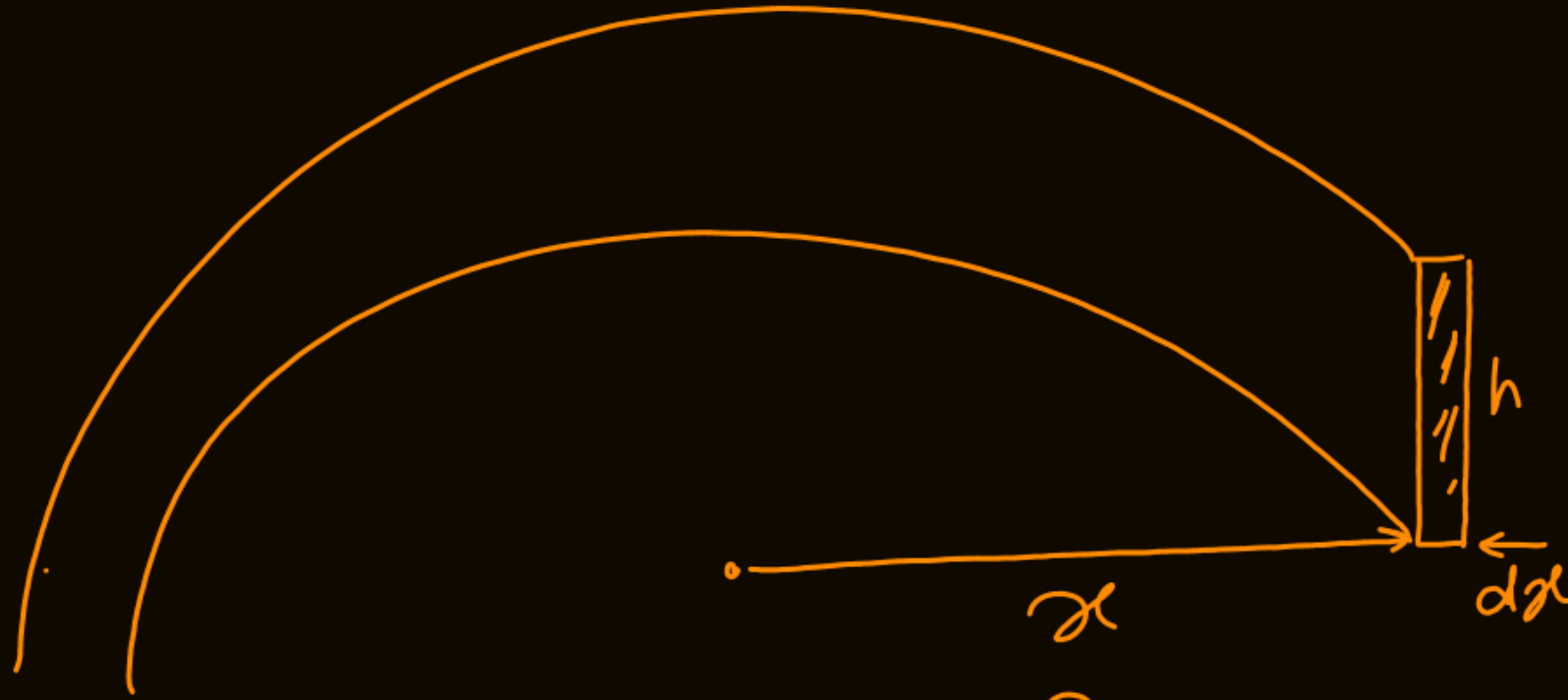
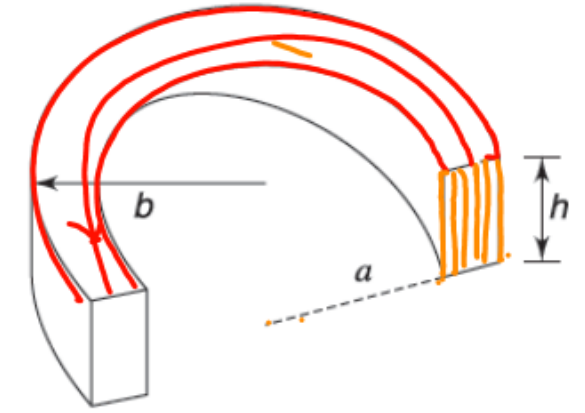
$$i = \frac{7.5}{10} = 1.5A$$

$$i = \frac{\mathcal{E}_1}{20} \times 1.5$$

$$\underline{10\Omega}$$

$$\mathcal{E}_1 = 20 \times 1.5 = \underline{\underline{150V}}$$

Q. 50: A conductor having resistivity ρ is bent in the shape of a half cylinder as shown in the figure. The inner and outer radii of the cylinder are a and b respectively and the height of the cylinder is h . A potential difference is applied across the two rectangular faces of the conductor. Calculate the resistance offered by the conductor.



$$dR = \frac{\rho \pi x}{h dx}$$

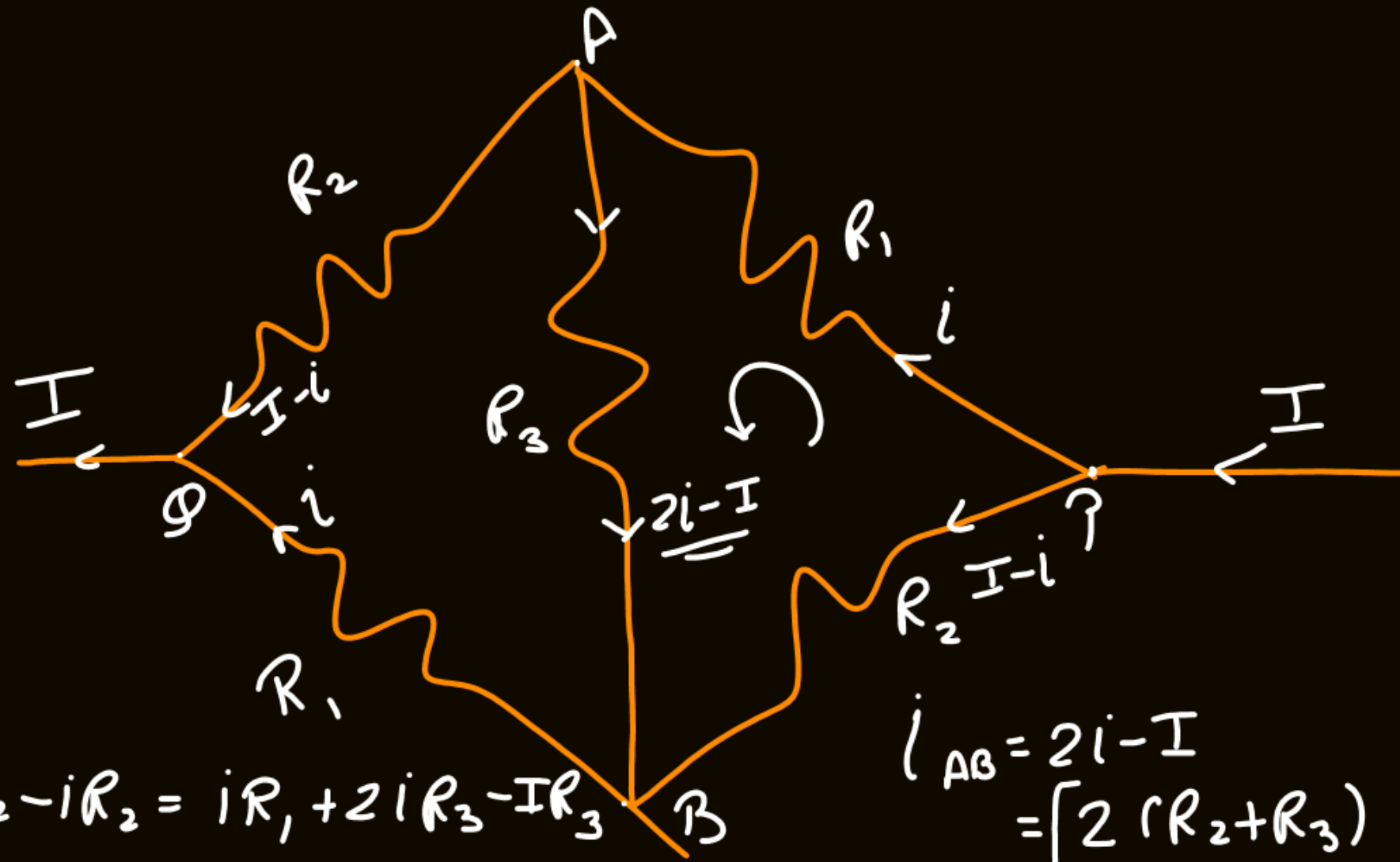
$$R_{eq} = \frac{\rho \pi}{h \ln b/a}$$

$$\frac{1}{R_{eq}} = \int \frac{1}{dR} = \int \frac{h dx}{\rho \pi x}$$

$$\frac{1}{R_{eq}} = \frac{h}{\rho \pi} \int_a^b \frac{dx}{x} = \frac{h}{\rho \pi} \ln b/a$$

$$(I-i)R_2 - iR_1 - (2i-I)R_3 = 0$$

$$i - (I-i) = 2i - I$$



$$IR_2 - iR_2 = iR_1 + 2iR_3 - IR_3$$

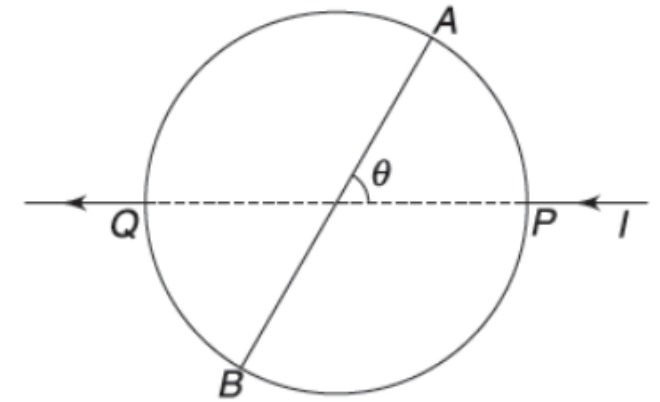
$$\frac{I(R_2 + R_3)}{R_1 + R_2 + 2R_3} = i$$

$$i_{AB} = 2i - I$$

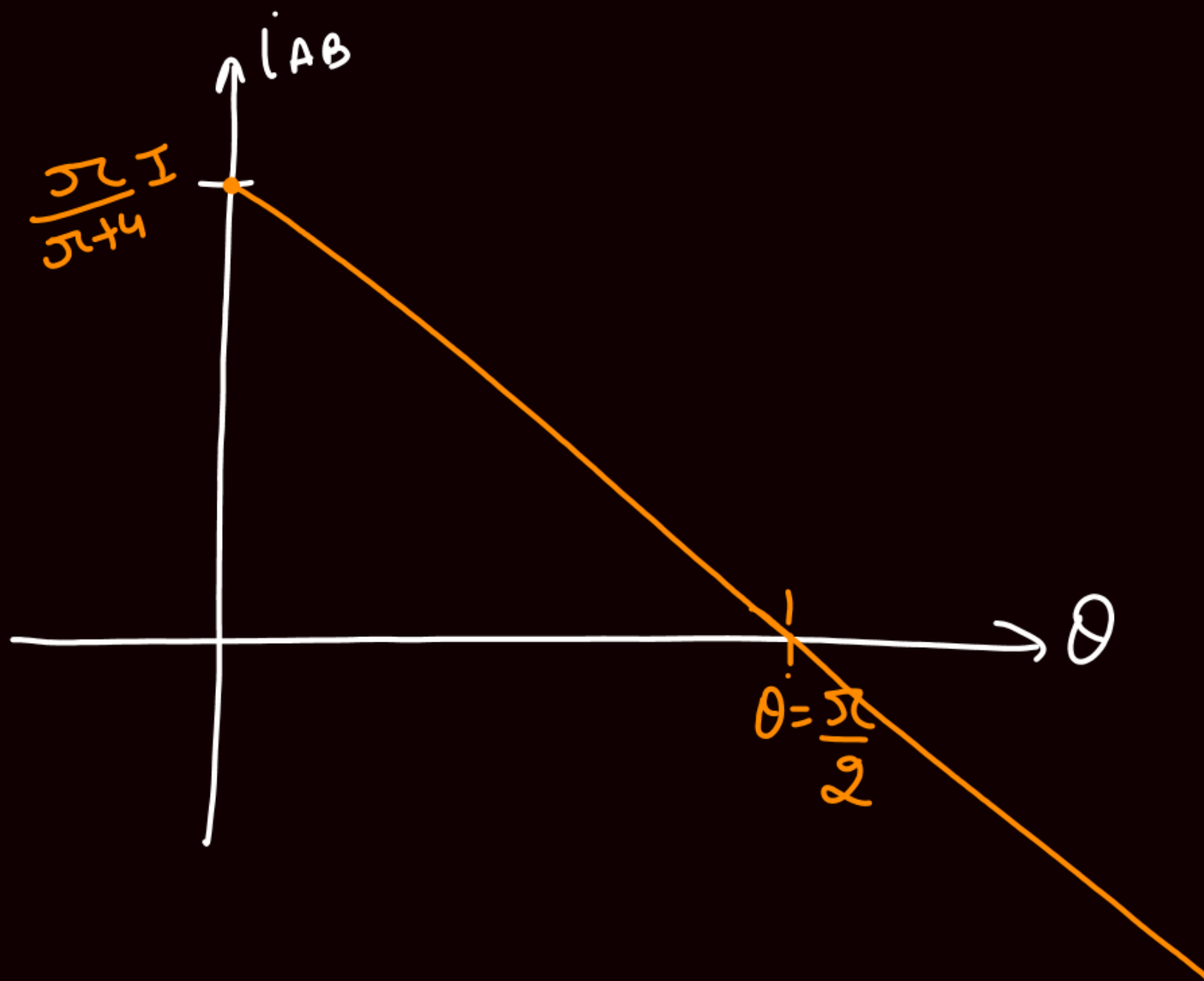
$$= \left[\frac{2(R_2 + R_3)}{R_1 + R_2 + 2R_3} - 1 \right] I = \frac{R_2 - R_1}{R_1 + R_2 + 2R_3} I$$

$$i_{AB} = \frac{\gamma(n-\theta) - \gamma\theta}{\gamma\theta + \gamma(n-\theta) + 2\gamma \times 2} I$$

Q. 62: A uniform conducting wire is in the shape of a circle. The same wire has been used to make its diagonal AB . A current I enters at point P and leaves at the diagonally opposite point Q . AB makes an angle θ with the line PQ . Find current (i), through AB as a function of θ . Plot a graph showing variation of i with θ ($0^\circ \leq \theta \leq 90^\circ$)



$$i_{AB} = \frac{\pi - 2\theta}{\pi + 4} \underline{I}$$



Sol:-

$$\frac{R_x}{R} = \frac{x}{100-x}$$

$$R_x = \frac{20}{80} \times 20\Omega = \underline{5\Omega}$$

$$R_x = x(100-x)^{-1}R$$

$$\ln R_x = \ln x - \ln(100-x) + \ln R$$

$$\frac{dR_x}{R_x} = \frac{dx}{x} + \frac{dx}{(100-x)} + \frac{dR}{R}$$

max^m error in R.

max^m possible

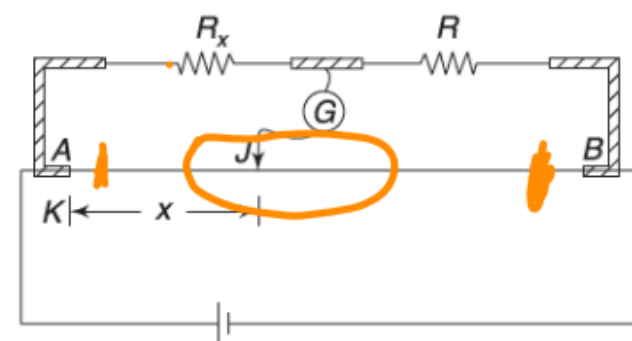
$$\text{error in } x \Rightarrow dx = 0.1\text{cm}$$

$$\frac{dR_x}{5\Omega} = \frac{0.1\text{cm}}{20\text{cm}} + \frac{0.1\text{cm}}{80\text{cm}} = \frac{0.5}{80} = \frac{dR_x}{R_x}$$

$$R_x = 5\Omega \pm 0.03\Omega$$

Q. 75: Figure shows an experimental set up to find the value of an unknown resistance (R_x) using a meter bridge. AB is the uniform meter bridge wire of length $L = 100$ cm. When the sliding jockey is placed at J ($AJ = x$), the galvanometer shows zero deflection. $AJ = x$ is known as balancing length and is measured using a scale having 1 mm as least count.

- In one experiment known resistance R was taken to be 20Ω and balancing length was measured as $x = (20.0 \pm 0.1)$ cm. Find the value of R_x .
- Show that fractional error in calculated value of R_x is least when $x = \frac{L}{2}$. What shall we do to ensure that x is close to $L/2$?



$$dR_x = \frac{0.5}{80} \times 5$$

$$= 0.03\Omega$$

$$\downarrow \frac{dR_x}{R_x} = \frac{100 dx}{x(100-x)} = \frac{100\text{cm} \times 0.1\text{cm}}{\underbrace{x(100-x)}} \uparrow$$

$$f(x) = x(100-x)$$

$$f'(x) = 100 - 2x = 0$$

$$x = 50\text{cm} = \frac{L}{2}$$



$$\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{2r} + \frac{1}{4r} + \frac{1}{8r} + \frac{1}{16r}$$

$$= \frac{1}{r} \frac{16 + 8 + 4 + 2 + 1}{16}$$

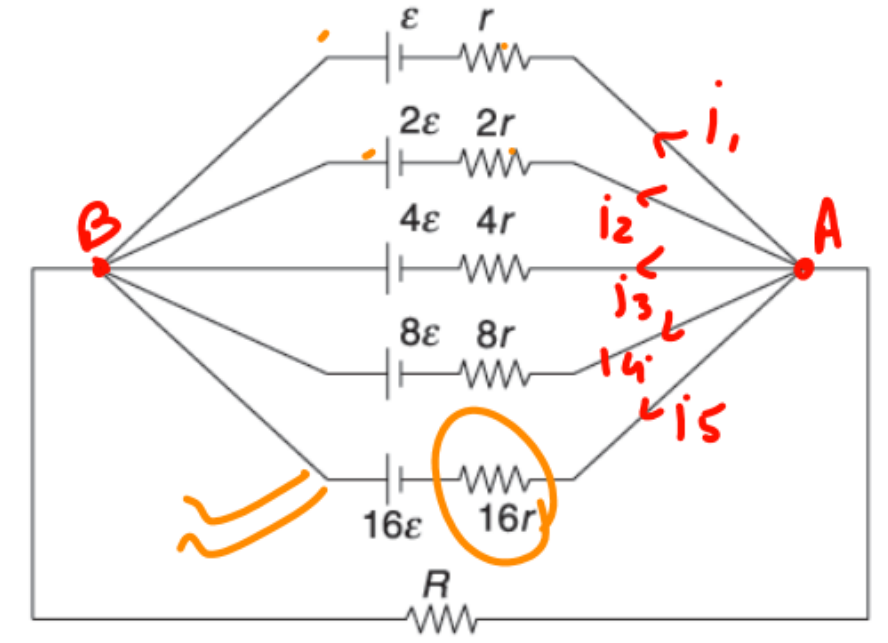
$$r_{eq} = \frac{16r}{31}$$

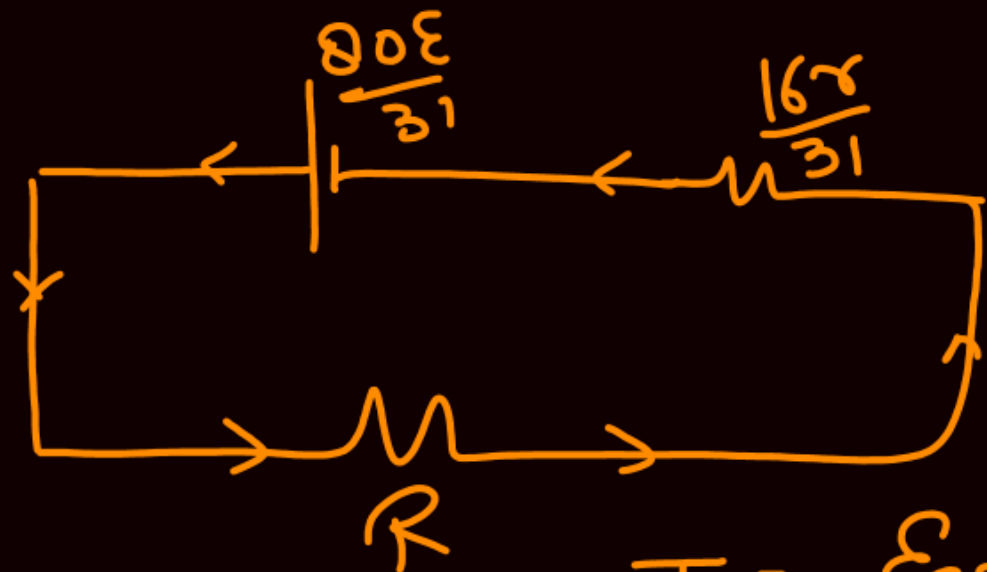
$$\mathcal{E}_{eq} = \frac{\frac{\mathcal{E}}{r} + \frac{\mathcal{E}}{2r} + \frac{\mathcal{E}}{4r} + \frac{\mathcal{E}}{8r} + \frac{\mathcal{E}}{16r}}{\frac{31}{16r}} = \frac{5\mathcal{E} \times 16}{31}$$

$$= \frac{80\mathcal{E}}{31}$$

Q. 77: Five cells have been connected in parallel to form a battery. The emf and internal resistances of the cells have been shown in figure. A load resistance R is connected to the battery.

- Which of the 5 cells will have maximum current flowing through it?
- Find the current through load resistance R .





$$\begin{aligned}
 I &= \frac{\mathcal{E}_{eq}}{R_{eq}} \\
 &= \frac{80\mathcal{E}}{31\left(R + \frac{16r}{31}\right)} \\
 &= \frac{80\mathcal{E}}{31R + 16r}
 \end{aligned}$$

$$V_B - V_A = \mathcal{E}_n - i\mathcal{r}_n$$

$$i\mathcal{r}_n = \mathcal{E}_n - (V_B - V_A)$$

$$i = \frac{\mathcal{E}_n - (V_B - V_A)}{\mathcal{r}_n}$$

$$\uparrow i = \underbrace{\left(\frac{\mathcal{E}_n}{\mathcal{r}_n} \right)}_{\text{same}} - \underbrace{\left(\frac{V_B - V_A}{\mathcal{r}_n} \right)}_{\substack{\uparrow \\ \mathcal{r}_n \uparrow \quad i \uparrow}}$$

↓